

House Price Prediction

Internship Project – Week 1

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Executive Summary

This project developed machine learning models to predict house prices using property features. The workflow included data exploration, preprocessing, visualization, model building, evaluation, and business recommendations.

Key Findings

- Area was the strongest predictor of house price.
- Bathrooms, parking, stories, and furnishing status significantly affected prices.
- Larger properties generally commanded higher market values.
- Correlation analysis showed strong positive relationships between area and price.

Model Performance

Model	MAE	RMSE	R ² Score
Linear Regression	9.700434e+05	1.324507e+06	0.652924
Random Forest	1.012055e+06	1.393128e+06	0.616030

- Linear Regression was used as a baseline model.
- Random Forest Regressor achieved better performance with higher R² and lower error metrics.
- Random Forest was selected as the final model.

Feature Importance

Top influential features:

1. Area
2. Bathrooms
3. Parking
4. Air Conditioning
5. Furnishing Status

Business Recommendation

Deploy a machine-learning pricing system using the Random Forest model to improve valuation accuracy and support data-driven decisions.

Conclusion

The project successfully developed a machine learning solution for house price prediction and demonstrated that ensemble models can provide accurate and reliable estimates for real-estate decision-making.

